1. Definition:

Pipeline architecture is a technique in which instruction execution is divided into a number of smaller stages. Different stages are executed simultaneously for different instructions using separate hardware units.

2. Working Principle:

- The **instruction execution** is divided into a sequence of **stages**.
- Each **stage** performs a part of the instruction.
- Multiple instructions are **overlapped in execution** in different stages.
- After the pipeline is full, **one instruction completes** in **every clock cycle**.

Example: 5-Stage Instruction Pipeline

IF Instruction Fetch → ID Instruction Decode & Register Fetch → EX Execute / ALU Operation → MEM Memory Access → WB Write Back

Clock Cycle →	1	2	3	4	5	6	7	8	9
Instruction 1	IF	ID	EX	MEM	WB				
Instruction 2		IF	ID	EX	MEM	WB			
Instruction 3			IF	ID	EX	MEM	WB		
Instruction 4				IF	ID	EX	MEM	WB	
Instruction 5					IF	ID	EX	MEM	WB

3. Benefits:

- Increases **throughput** of the CPU.
- Higher **CPU utilization**.
- Improves **performance**.
- Better **speedup** with multiple instructions.

4. Limitations:

- Does not reduce **latency** of a single instruction.
- **Pipeline hazards** (data, control, structural) reduce efficiency.
- More **hardware complexity**.
- Performance decreases if stages are **unbalanced**.

Q. Pipeline Hazards: definition, types (Structural, Data, Control) & explanation

Ans:

Definition: A hazard in pipelining is a condition that prevents the next instruction in the pipeline from executing during its designated clock cycle.

Types of Pipeline Hazards:

There are three main types of pipeline hazards:

1. Structural Hazards

- Occur when hardware resources are insufficient for the pipeline stages.
- Two or more stages need the same resource in the same clock cycle.
- **Example:** Single memory used for both instruction fetch and data access.

2. Data Hazards

- Occur due to data dependencies between instructions.
- An instruction depends on the result of a previous instruction still in the pipeline.
- **Types:** RAW (Read After Write), WAR (Write After Read), WAW (Write After Write).
- **Handled by:** Forwarding, stalling (bubbles).

3. Control Hazards

- Occur due to change in the flow of control.
- Caused by branch, jump or other control instructions.
- Next instruction address is unknown until the branch is resolved.
- **Handled by:** Branch prediction, stalling, flushing.

Pipeline Stages (Example):

IF → ID → EX → MEM → WB

IF	ID	EX	MEM	WB
Instruction Fetch	Instruction Decode	Execute	Memory Access	Write Back

> Hazards reduce the efficiency of pipelining. They must be detected and handled to maintain pipeline performance.